

■ Distribute

`Distribute[g[x1, x2, ...]]` distributes g over Plus appearing in any of the x_i .

`Distribute[expr, f]` distributes over f .

`Distribute[expr, f, g]` performs the distribution only if the head of $expr$ is g .

`Distribute` effectively implements the distributive law for operators f and g . ■ `Distribute` explicitly constructs the complete result of a distribution; `Expand`, on the other hand, builds up results iteratively, simplifying at each stage.

■ Example: `Distribute[g[a+b,c+d]]` $\longrightarrow$ `g[a, c] + g[a, d] + g[b, c] + g[b, d]`.

■ `Distribute[g[a+b,f[x,y],c], f]` $\longrightarrow$ `f[g[a + b, x, c], g[a + b, y, c]]`. ■ See page 250. ■ See also: `Expand`, `Thread`.